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Anonymous Author(s)
Affiliation
Address
email

Abstract

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1 Introduction

[This is only a skeleton of the intro] Much of the world knowledge can be expressed in terms of relations, but oftentimes such relations are graded, implied, and nonliteral. Such kind of implicit inference of event relations is common in human language processing... (cite) Talk about eliciture and why its implied relationships between any two events are non-mandated. (cite)

Previous studies have attempted to show how LMs can build a world model through pragmatic inference... (cite) To the best of our knowledge, Pan & Kehler is the only work that examines LM's ability to perform eliciture inference. They demonstrate that LMs not only draw eliciture but also leverage eliciture to guide downstream sentential processing. we aim to extend on their findings by evaluating models of different families and studying the internal representations of implicit causal inference.

The benefits of understanding elicitors in LMs are multifold: 1. whether LMs–trained with distributional objective–demonstrate human-like nonliteral inference in non-mandated setting 2. more comprehensive views into how LMs build an internal world model 3. Potentially understand the implicit bias in LMs, which has implications in safety and alignment

To these aims, we explore three main questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Background

3 Experiments

3.1 Measuring Implicit Causal Inference through Model Behaviors

Models. In this and all of the following experiments, we test Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024] base checkpoints.

Dataset. We use 150 manually authored pairs of sentences. In each pair, the modifier or relative clause in the critical sentence provides a plausible cause of that outcome based on physical or everyday world knowledge; in the *control* sentence, it provides an incidental detail. For example, *The executive who falsified quarterly revenue figures was fired* is paired with *The executive who wore a blue silk tie*

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was fired. This dataset extends the no-context eliciture paradigm of Pan and Kehler [2025], which tests whether an apparently descriptive modifier is nevertheless understood as an explanation.

Yes/No. We first present the critical/control sentences as context and then ask, "Does [modifier content] explain why [outcome]?" We compute the probabilities of the complete answers *Yes* and *No*, including capitalization variants, and normalize them to obtain $P(\text{Yes})$.

7-point Likert. We first present the critical/control sentences as context and then ask how strongly the modifier explains the outcome on a scale from 1 (definitely not) through 4 (uncertain) to 7 (definitely yes). We compute a probability for each of the seven complete answers and use their probability-weighted mean as the model's rating.

"I don't know why." Following Pan and Kehler [2025], we remove the sentence-final period and append *, and I don't know why.* We use the surprisal of this continuation—its negative summed log probability—as the score. If the modifier has already supplied an explanation, subsequently claiming not to know why should be less expected and therefore more surprising.

For all three measures, paired accuracy is the proportion of pairs in which the critical sentence scores strictly above its control; chance is .5, ties count as incorrect.

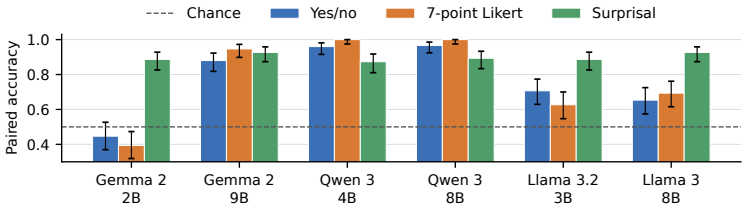


Figure 1: Paired accuracy on 150 pairs of implicit causal inference items with 95% CIs, **Models demonstrate implicit causal inference most consistently through probability.** Not all models can faithfully report their implicit causal inference through Yes/No and 7-Point Likert rating. The performance of metalinguistic causal inference increases with model size.

Results.

3.2 Identifying Internal Representations of Implicit Causality Inference

Pragmatic theories state that cross-sentence causal inferences should be much stronger, and are usually non-optional in human communication. This motivates RQ2: Do LMs form internal representations of implicit causal relationships across sentences? Specifically, we want to ask whether model activations distinguish causal relationships from other relations. If a representation is specific to causality, a causal direction should separate causal pairs well but transfer less reliably to temporal or plausibility pairs.

Datasets.

CAA directions. Contrastive Activation Addition (CAA) represents a relation with a direction in a model's activation space [Rimsky et al., 2024]. At each tested layer, we subtract the final-token activation of every negative item from that of its matched positive item, then average these differences over the training pairs. An evaluation item receives a scalar score by projection onto this direction of averaged difference; paired accuracy is the proportion of evaluation pairs for which the positive item scores higher.

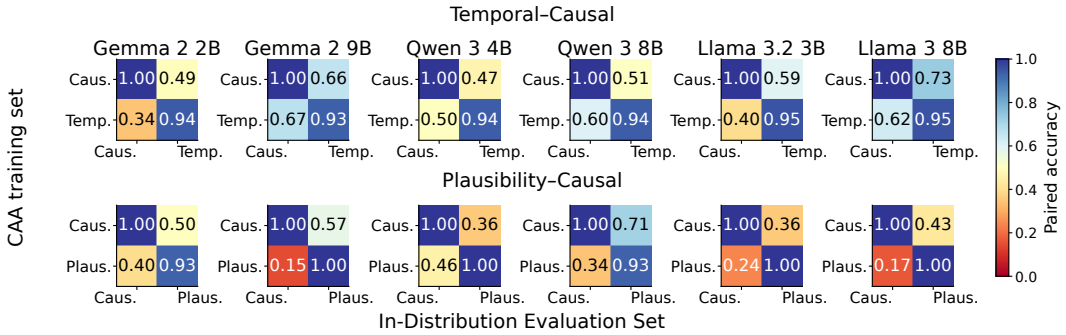


Figure 2: Cross-transfer paired accuracy between causal and temporal relations (top row) and causal and plausibility relations (bottom row). Within each heatmap, rows show the CAA training set and columns show the in-distribution evaluation set. **Causal CAA directions are largely relation-selective and distinct from temporal and plausibility directions.**

Results.

3.3 Generalization to Naturalistic Scenarios

Dataset.

Table 1: Mean paired differences in causal CAA projection in naturalistic causal scenarios. Positive values indicate that a demographic group has a greater mean projection than the white-male condition. Bolded values indicate statsitically significant difference from the white-male projection scores. **Demographic descriptions are associated with differences in causal CAA projections.** Minority females generally have larger projection raw mean scores than minority males along the implicit causal directions.

	Gemma 2 2B	Gemma 2 9B	Qwen3 4B	Qwen3 8B	Llama 3.2 3B	Llama 3 8B
Minority male − White male	**.41 ± .32**	**1.9 ± .40**	.073 ± .093	**.29 ± .17**	.025 ± .029	**.030 ± .020**
Minority female − White male	**.57 ± .32**	**2.3 ± .41**	**.15 ± .10**	**.39 ± .15**	**.069 ± .033**	**.060 ± .021**

Results.

4 Discussion

References

Abhimanyu Dubey et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL https://arxiv.org/abs/2407.21783.

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A Testing the Generalization to Within-Sentence Implicit Causal Inference

We now ask whether the learned directions remain informative on unseen causal constructions within a single sentence (eliciture).

Training and evaluation data.

Analysis.

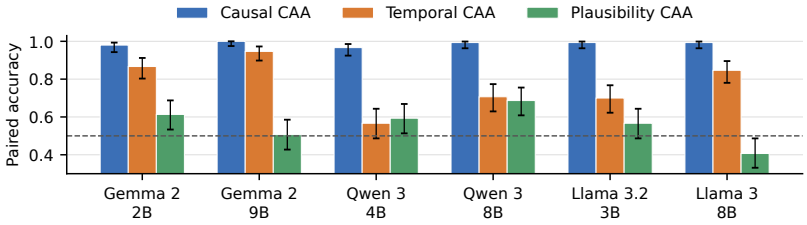


Figure 3: Paired accuracy of causal, temporal, and plausibility CAA directions on 150 unseen within-sentence causal inference pairs with 95% CIs. **Causal CAA directions generalize consistently to within-sentence implicit causal inference.** Temporal and plausibility CAA directions transfer less consistently to within-sentence implicit causal items. The dashed line denotes chance.

Results.

B Raw CAA Projection Distributions

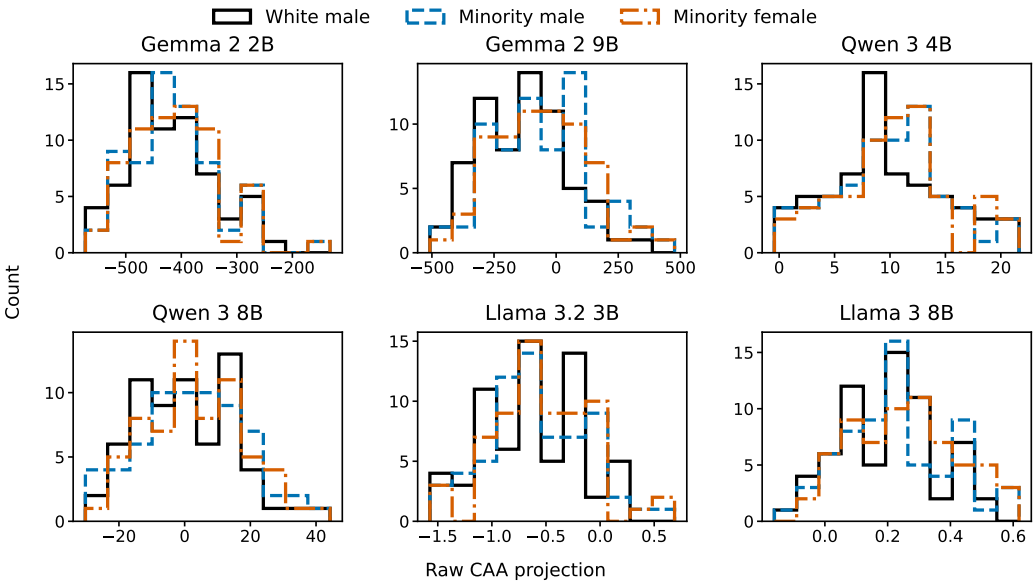


Figure 4: Raw causal CAA projection distributions for three matched demographic conditions across 65 scenarios per model. **The three conditions have substantially overlapping projection distributions in every model.** Colors and line styles distinguish the White-male, minority-male, and minority-female conditions.